POLYNOMIALS, FACTORIALS AND A CONTROLLED ERROR

Power series: build e^{1+x}

Start with a value and a tangent. Add powers to make the curve agree more closely.

1 / WHAT IS BEING EXPANDED?

The small variable measures distance from a known point

A **power series** has the form $a_0 + a_1x + a_2x^2 + \cdots$, with fixed coefficients a_n. Stopping after x^N gives a polynomial approximation.
For $F(x) = \mathrm{e}^{1+x}$, the offset is $x = t - 1$: $x = 0$ **means the exponential's argument is** $t = 1$, where the value is e.



argument $t = 1 + x$

2 / MAKE THE POLYNOMIAL AGREE

Why do factorials appear?

The exponential is its own derivative: $F' = F$, so $F(0) = F'(0) = F''(0) = \cdots = \mathrm{e}$. Start with a cubic,

$$P_3(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

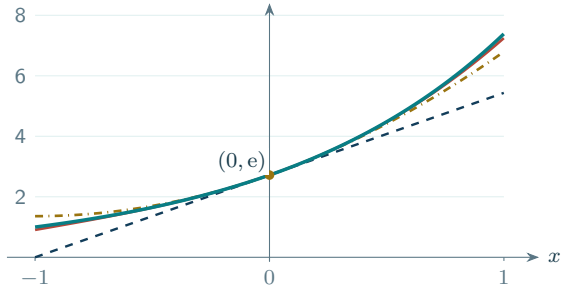
Match the value, slope and higher derivatives at 0:

match	equation	coefficient
value	$a_0 = \mathrm{e}$	$a_0 = \mathrm{e}$
slope	$a_1 = \mathrm{e}$	$a_1 = \mathrm{e}$
second derivative	$2!a_2 = \mathrm{e}$	$a_2 = \mathrm{e}/2!$
third derivative	$3!a_3 = \mathrm{e}$	$a_3 = \mathrm{e}/3!$

Differentiating a_nx^n exactly n times gives $n!a_n$. Thus the matching coefficient is $a_n = \mathrm{e}/n!$. Here $n! = 1 \cdot 2 \cdots n$ and $0! = 1$.

3 / SEE THE SUCCESSIVE APPROXIMATIONS

More powers, more local agreement



e^{1+x} P_1 (line) P_2 (quadratic) P_3 (cubic)

Each curve shares the value and slope at $x = 0$. The quadratic also matches the second derivative; the cubic matches the third as well.

4 / CONTINUE THE CONSTRUCTION

A Taylor series is a power series about a chosen point

$$\mathrm{e}^{1+x} = \mathrm{e}\sum_{n=0}^{\infty}\frac{x^n}{n!} = \mathrm{e}\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right).$$

In the original variable, $\mathrm{e}^t = \mathrm{e}[1 + (t-1) + (t-1)^2/2! + \cdots]$: powers of the distance from $t = 1$. Dividing by e recovers the familiar series for e^x.

5 / USE A SMALL OFFSET

Estimate $\mathrm{e}^{1.1}$ with $x = 0.1$

Use the partial sum $P_N(x) = \mathrm{e}\sum_{n=0}^{N} x^n/n!$. Each next term is the current term multiplied by $x/(n+1)$, where n is the current power.

degree N	$P_N(0.1)$
0	2.718281828
1	2.990110011
2	3.003701420
3	3.004154467
4	3.004165794

For comparison, $\mathrm{e}^{1.1} = 3.004166023946\ldots$. The finite sum approximates the value; the infinite series equals it.

6 / WHY DOES THE INFINITE SUM WORK?

Control what has been left out

Matching derivatives suggests the series. Equality needs the remainder $R_N(x) = \mathrm{e}^{1+x} - P_N(x)$ to tend to zero. Taylor's remainder theorem gives, for $|x| \le r$ with $r > 0$,

$$|R_N(x)| \le \mathrm{e}^{1+r}\frac{|x|^{N+1}}{(N+1)!}.$$

Successive bounds have ratio $|x|/(N+2) \to 0$, so the error tends to zero for every fixed x. For $x = 0.1$, using $r = 1$ gives

$$|R_4(0.1)| \le \frac{\mathrm{e}^2(0.1)^5}{5!} < 7 \times 10^{-7}.$$

Next: build a logarithm. Integrating a power raises its exponent and divides by the new exponent: $\int_0^x u^n\,du = x^{n+1}/(n+1)$. Applying this to the geometric series for $1/(1+u)$ produces $\ln(1+x)$ and, after checking the endpoint, the alternating series for $\ln 2$.

Reference. Exponential power series: NIST DLMF, equation 4.2.19. All variables on this sheet are real; N is a non-negative integer.

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